Past exam 2021

Wednesday, 22 February 2023 1:39 PM

Q1] $2A_1 + 3A_2 \rightarrow 2A_3$ No residual amount of A_1, A_2 & A_3

$A_1 + A_2 + A_3 \rightarrow 3P$

$A_1 + \frac{3}{2}A_2 \rightarrow A_3$

$A_1 + A_2 + A_3 \rightarrow 3P$

$A_1 + A_2 + (A_1 + \frac{3}{2}A_2) \rightarrow 3P$

$2A_1 + \frac{5}{2}A_2 \rightarrow 3P$ $\quad \frac{A_1}{A_2} = \frac{2}{5/2} = \frac{4}{5}$

$2A_1 + 3A_2 - 2A_3 = 0$

$A_1 + A_2 + A_3 - P = 0$

$$\begin{bmatrix} 2 & 3 & -2 & 0 \\ 1 & 1 & 1 & -1 \end{bmatrix}$$

Solution:

$dN_j = \alpha_j dX$ $\quad X \rightarrow$ extent of reaction $\}$ for 1 reaction

$\Delta N_1 = N_{1f} - N_{1i} = -2X_1 - X_2$ $\quad X = \frac{N_f - N_i}{\alpha}$

$\Delta N_2 = -3X_1 - X_2$

$\Delta N_3 = 2X_1 - X_2$

$\Delta N_4 = 3X_2$

$\underline{\Delta N} = \underline{\underline{A}}\,\underline{X}$

$$\Delta \begin{bmatrix} N_1 \\ N_2 \\ N_3 \\ N_4 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ -3 & -1 \\ 2 & -1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$\underline{\underline{A}}^T = \begin{bmatrix} -2 & -3 & 2 & 0 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$ Wide matrix (max rank = 2) — Has a null space

$\underline{\underline{A}}^T z = 0$

So $\underline{\underline{A}}^T z = 0$ → must also have a non-trivial solution

$$z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}$$

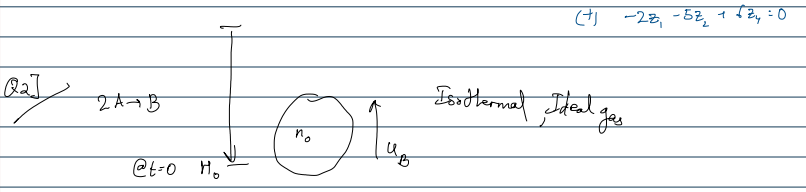
$\underline{\underline{A}}^T z = 0$ $\quad -2z_1 - 3z_2 + 2z_3 + 0z_4 = 0$ $\quad -2z_1 - 3z_2 + 2z_3 = 0$

$-1z_1 - z_2 - z_3 + 3z_4 = 0$ $\}(-1)$ $\quad 2z_1 + 2z_2 - 2z_3 + 6z_4 = 0$

$-z_2 + 4z_3 - 6z_4 = 0$

$(+1) \quad -2z_1 - 5z_2 + 6z_4 = 0$

Q2]



→ Bubble volume at time t.

(a) Production of gas → rate $\quad r = k[A]^2$

$V = \frac{NRT}{P}$

Pressure balance: $P = \rho g h_0$ $\quad P = \rho g (u_B \cdot t)$

H_0 | 0 $\quad (H_0 - u_B t)\rho g + P_a = P$

Mass balance: $dN_j = \alpha_j dX$

Mass balance : $dN_j = \alpha_j \, dx$

$$\frac{dN_j}{dt} = \alpha_j \frac{dx}{dt} = \alpha_j V \cdot r \qquad r = k[A]^2$$

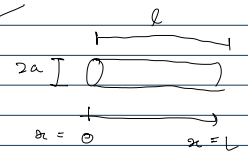
$$\frac{dN_A}{dt} = -2V \cdot r \qquad V = \frac{NRT}{P}$$

$V(t)$

$$\frac{dN_A}{dt} = -2\frac{NRT}{P} r$$

$$\alpha_j \frac{dx}{dt} = \alpha_j V \cdot r$$

Q3]



$\hat{V}_{max}$ = max. area-specific reaction rate

pore-diffusion problem

(i) Shell balance :

$$D\frac{d^2c}{dx^2} + \frac{P}{A}\hat{r}(c) = 0$$

BC: @ $x = 0$, $c = c_s$

@ $x = l$, $\frac{dc}{dx} = 0$

$$\frac{l \cdot 2\pi a}{\pi a^2} = \frac{2l}{a}$$

$$D\frac{d^2(c_s u)}{d(\rho l)^2} = -\hat{r}(c) \cdot \frac{2l}{a}$$

$$\hat{r}(c) = R(u)\,\hat{r}(c_s)$$

$$\frac{Dc_s}{l^2}\frac{d^2u}{d\rho^2} = -\hat{r}(c) \cdot \frac{2l}{a}$$

$$\frac{d^2u}{d\rho^2} = -R(u)\frac{\hat{r}(c_s)}{Dc_s} l^2 \cdot 2 \cdot l$$

$$\frac{d^2u}{d\rho^2} = -\left(\frac{2l^2 \hat{r}(c_s)}{a\,D_{c_s}}\right) \cdot R(u) \cdot l$$

$$\frac{d^2u}{d\rho} = -\phi^2 \cdot R(u)\, l$$

$$R(u) = \frac{\hat{r}(c_s)}{\hat{r}(c_s)} = \frac{\frac{\hat{V}_{max} c}{K + c}}{\frac{\hat{V}_{max} c_s}{K + c_s}} = \frac{c}{K+c}\frac{(K+c_s)}{c_s} = \frac{c}{c_s}\frac{(K+c_s)}{(K+c)}$$

$$= \frac{u(K+c_s)}{(K+c)} = \frac{u}{c_s}\frac{(K+c_s)}{(K+c)}$$

$$= \frac{u}{c_s}\frac{(K'+1)}{\left(K' + \frac{c}{c_s}\right)}$$

$$= \frac{u(K'+1)}{(K'+u)}$$